Business Intelligence 381 | Milestone 4

Unemployment and Labour Market Exclusion in South Africa

Final Project Report: Socioeconomic Development and Household Well-being in South Africa (SDHSA)

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Companion files

Power BI dashboard (Dashboard.pbix); executive presentation; final ethics report and responsible AI usage log; handover and project review; R scripts and Shiny app.

1. Business problem and project goals

Unemployment is South Africa's most stubborn socioeconomic problem. According to Statistics South Africa (2024a, 2024b, 2024c, 2024d), the official unemployment rate was 32.9% in the first quarter of 2024, 33.5% in the second, 32.1% in the third, and 31.9% in the fourth, with roughly 8.2 million people out of work at the start of the year. But a single national figure masks wide differences between provinces, age groups, and communities. Those differences are exactly what the Department of Employment and Labour (DEL) and provincial economic development departments need when deciding where to spend limited programme budgets.

The decision problem is therefore: which demographic and geographic groups face the highest risk of unemployment, and which factors are associated with that risk? In Milestone 3, we narrowed this into a single analytical question: to what extent can demographic, regional, and educational indicators predict an individual's risk of unemployment, and which predictors matter most? We worked through this question using CRISP-DM (Chapman et al., 2000), business understanding, data understanding, data preparation, modelling, evaluation, and deployment (Milestone 1 covered the first two phases, Milestone 2 the preparation phase, Milestone 3 the modelling, evaluation, and Shiny app, and Milestone 4 the dashboard and final synthesis).

Milestone 1 set five business objectives. Table 1 shows where each one is answered in this report and what we found. Objective 4 is the one we could not deliver, and we say why.

Business objective (Milestone 1)	Answered in	Result
1. Identify the groups most affected, by age and gender	Section 3	Youth 45.6% against 23.0% for adults; young women 49.4%, young men 42.3%
2. Compare provinces with the national picture	Section 3	Five of nine provinces above the national 32.6%; North West highest (40.0%), Western Cape lowest (20.7%)
3. Examine education and unemployment	Section 3	Tertiary 20.0% against 37.0% for primary or lower; matric holders are 66.2% of the unemployed
4. Examine sectors and formal or informal work	Sections 2 and 8	Not delivered. Sector fields leak the outcome, and the formal/informal coding looks swapped
5. Clear reporting for non-technical users	Sections 3 and 5	Power BI dashboard; Shiny app with plain-language results

Table 1. Business objectives and outcomes. The technical criteria also held: documented data-quality fixes, weighted rates within 0.1 percentage point of Stats SA, more than three segmentation dimensions, and a scripted seeded pipeline.

2. Data strategy, preparation and integration

What we used. Before choosing any data, we inspected all eleven QLFS-derived extracts (D01 to D11). The household files covering dwellings, services, energy, connectivity, and transport (D04, D05, D06, D08, and D10) have no plausible link to labour status, so we excluded them. We also left out D07 (hunger) and D09 (income and grants) to keep the scope focused. Four files carried the analysis: D11 (labour market, all four 2024 quarters) as the backbone, and D01, D02, and D03 (population, learning and economic participation, first quarter only) for richer detail.

The integration decision that mattered most. We tested, rather than assumed, whether D11 could be joined to D01 to D03. It cannot: not one person key from D01 appears in D11, even though 938 of 1,037 area codes overlap. The files come from the same survey areas but different dwellings. We therefore ran two parallel tracks. Track A uses D11 alone and feeds the models, the weighted analysis, and the dashboard. Track B joins D01, D02, and D03 on household and person number (45,326 rows before the inner join, 45,322 after) and provides descriptive context only.

Status Label	Rows	Approx. Share
Employed	63,390	38.4%
Unemployed	32,043	18.8%
Other Inactive	59,445	34.9%
Discouraged Seeker	13,499	7.9%
Total	170,377	100%

Table 2a. Track A status composition (n = 170,377). The model target is Unemployed (1) vs all other statuses combined (0)

Source	Raw rows	Cleaned rows	Removed	What was removed
D01	70,440	45,326	25,114	Outside age 15 to 64 (25,100); repeated composite IDs (14)
D02	70,462	70,440	22	Exact duplicate rows
D03	70,460	70,440	20	Exact duplicate rows; salary values above 1e300 set to missing
D11 (Track A)	170,432	170,377	55	Age outside 0 to 100 (25); repeated person-quarter IDs (30)

Table 2b. Row counts before and after cleaning. D11 keeps repeated people across quarters, so its key is person plus quarter.

Table 3 lists the preparation decisions that changed the final results, each with the problem, the R treatment, the effect, and how we checked it.

Problem	Treatment in R	Effect	Verification
18-digit IDs rounded when read as numbers, so joins found no matches	Read IDs as text; build key household_person_year_quarter	Keys survive intact; one-to-one joins possible	Zero duplicate keys in the cleaned tables
D11 uses 1.8e308 as "not applicable" (up to 87% of some columns)	Recode every value above 1e300 to NA; set negative amounts and hours to NA	Averages and models no longer distorted	Count of values above 1e300 is 0 after cleaning
Missing hours worked	Median within occupation-code groups; all-missing groups stay missing	No rows lost; descriptive use only	Row counts unchanged by imputation
Province spelled in 33 variants	Trim and standardise before encoding (02b_fix_baked.R)	Nine provinces, not 33 dummy columns	Province dummy count checked after baking
Status held as a number	Recode to Employed, Unemployed, Discouraged Seeker, Other Inactive	Traceable labels	Cross-tab of code against label, none unmapped
Employed-only fields (occupation, industry, informal status, earnings, hours) and outcome fields would leak the answer	Drop them, plus identifiers and survey weights, before modelling	Predictions use only characteristics known before labour status	Leakage list documented; recipe fitted on training data only
Only 18.8% of records are unemployed	Stratified 70/30 split (seed 2026); thresholds tuned instead of discarding data	Training 119,263 / test 51,114, prevalence preserved	Class share checked in both sets

Table 3. Key preparation decisions with before-and-after evidence.

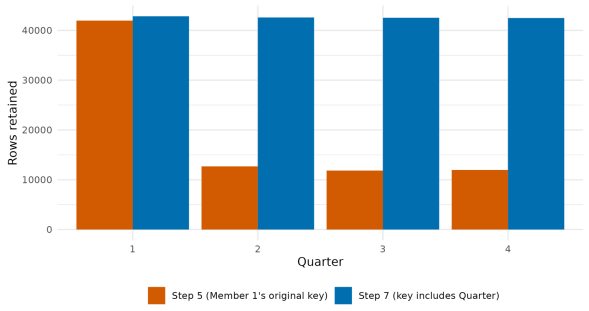


Figure 1a. Rows kept per quarter before (orange) and after (blue) adding Quarter to the duplicate key (Member 2).

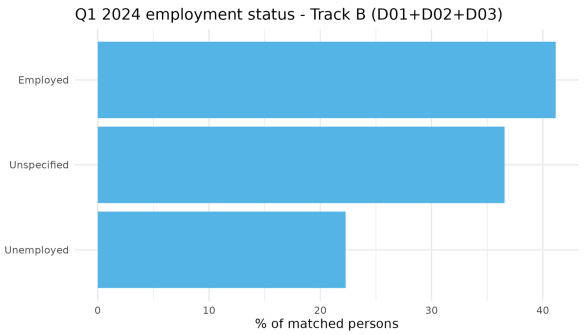


Figure 1b. Track B employment status, Q1 2024 (Member 2).

The final dataset. Track A has 170,377 person-quarter records and 18 columns: Employed 65,390 (38.4%), Unemployed 32,043 (18.8%), Other inactive 59,445 (34.9%), and Discouraged seeker 13,499 (7.9%). The derived fields include youth flag (15 to 34), age band, education group, geographic type, quarter label, and the model outcome (1 = unemployed; 0 = everything else). For the weighted analysis we added the survey weight and education from the raw D11 file, and we joined one-to-one with row counts checked (170,377 in, 170,377 out). The data dictionary is in Appendix A.

3. What the data shows

Two views of the same 170,377 records are used in this report, and it matters which one a number comes from. The weighted view applies the survey weight, so each record counts as the number of people it represents. It estimates the national picture and is the view we compare with Stats SA. The sample view, which the Power BI dashboard shows, counts records as they are. Unemployment follows the official definition in both: unemployed divided by employed plus unemployed, for people aged 15 to 64. Weighted, the pooled 2024 rate is 32.6%, rising to 40.4% when discouraged work-seekers are added, and about 8.15 million people were unemployed in an average quarter.

Accuracy. Our weighted estimates are 32.9%, 33.5%, 32.1% and 31.9% for Q1 to Q4, within 0.1 percentage point of the published figures. That is why we trust the group comparisons that follow. Without weights the pooled rate is 32.9% rather than 32.6%, and provincial figures shift by up to 1.4 points, so the dashboard numbers in Table 4 are close to, but not the same as, the national estimates.

Age is the strongest dividing line. Young people (15 to 34) have a rate of 45.6%, double that of people aged 35 and over (23.0%), and they make up 59.5% of all unemployed people. Among 15 to 24 year olds the rate climbs to 60.1%. Women are more exposed than men (34.8% against 30.7%), and the gap is concentrated among the young: 49.4% for young women against 42.3% for young men. The dashboard shows the same pattern in the sample, and its youth NEET rate (not in employment, education or training) is 42.1%.

Where, in rates and in numbers. North West (40.0%), Eastern Cape (39.3%) and Free State (37.4%) have the highest rates, and the Western Cape the lowest (20.7%). Five of the nine provinces sit above the national rate. Rates and headcounts tell different stories, though. Gauteng has a mid-table rate (34.4%) yet holds 32.6% of all unemployed people, followed by KwaZulu-Natal at 15.2%. Planning on rates alone would miss where most people actually are.

Education and area. People with tertiary qualifications have a rate of 20.0%, against 37.0% for both the primary-or-lower and the secondary groups. For planners the volume matters more: people whose highest qualification is secondary school or matric make up 66.2% of the unemployed. Rural traditional areas have the highest area rate (42.9%) and hold 29.2% of the unemployed. A disengagement proxy built on the Track B sample, covering people who are not studying, working nor looking for work, points the same way: about 30% in rural traditional areas against 18% on farms and in urban areas, and 26.9% for adults against 16.3% for youth (Milestone 2; students are not counted as disengaged, so the youth figure is understated). These are associations; they do not show that education or location causes unemployment.

The Power BI dashboard

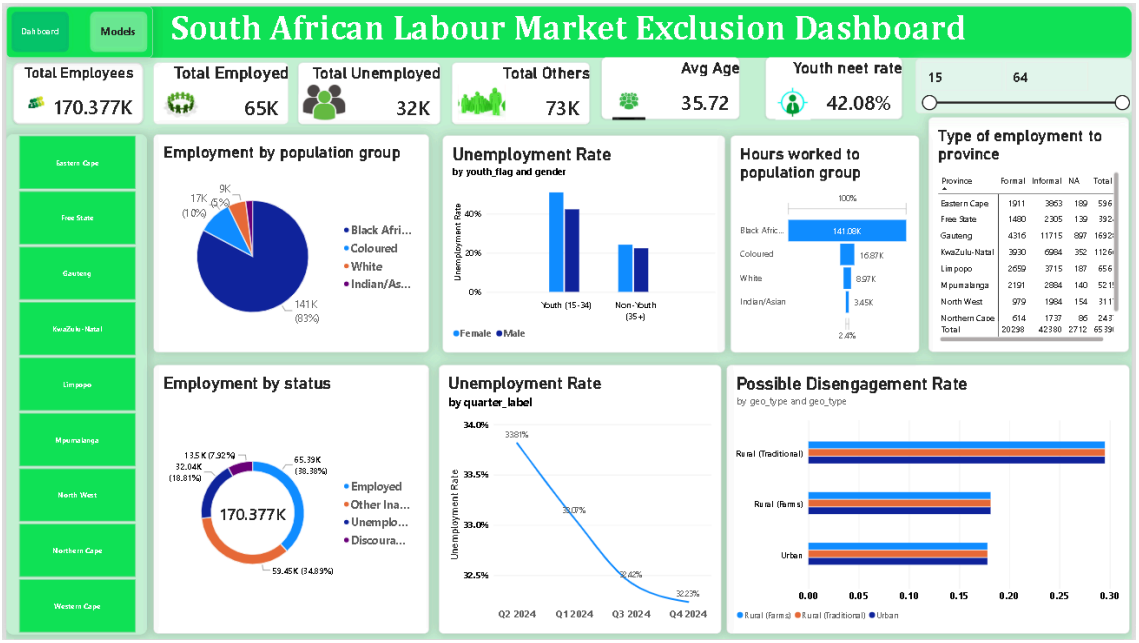


Figure 2. Dashboard page 1 : KPI row, province selector, age slider. and linked charts, with all provinces selected.

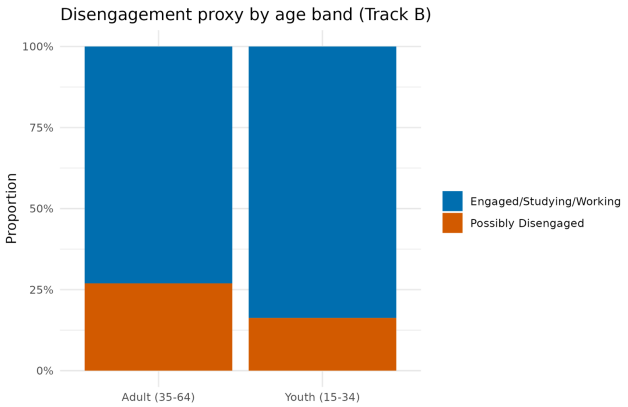


Figure 3. Disengagement proxy by age band, Track B (Member 2, Milestone 3): about 27% of adults against 16% of youth are possibly disengaged.

Design. The dashboard answers the decision questions on one page. The KPI row (records, employed, unemployed, others, average age, and youth NEET rate) gives the summary. The province buttons and the age slider filter every chart. The status donut and the population group charts show who is active in the economy and who is not. The youth and gender chart shows whether youth status and gender compound exclusion. The quarterly line shows the trend, and the disengagement chart shows where drop-out concentrates. A second page, Models, carries the ROC comparison and the top predictors. The green theme and white tiles separate sections and lead the eye from summary to detail. It draws on a single dataset, because the extracts could not be joined without inflating counts.

Measure	Dashboard (sample records)	Weighted estimate	Difference
Pooled unemployment rate	32.9%	32.6%	0.3 pt
Rate, Q1 / Q2 / Q3 / Q4	33.1 / 33.8 / 32.4 / 32.2%	32.9 / 33.5 / 32.1 / 31.9%	0.2 to 0.4 pt
Youth (15 to 34) rate	46.3%	45.6%	0.7 pt
People unemployed	32,043 records	8.15 million a quarter	Different units

Table 4. The dashboard (sample view) against the weighted estimates. Read the dashboard for patterns and rankings; quote national levels from the weighted figures.

4. Modelling and evaluation

Task and candidates. We framed the problem as supervised binary classification: is a person unemployed (1) or not (0, which includes employed, discouraged and inactive people)? There are 32,043 positives (18.81%) and 138,334 negatives. Three models cover different needs: logistic regression as an interpretable baseline, Naive Bayes as a probabilistic alternative, and a random forest (ranger, 500 trees, tuned over mtry and minimum node size with 5-fold cross-validation) for non-linear patterns, plus class-weighted and original forest variants. We used a stratified 70/30 split with seed 2026, and fitted the preprocessing recipe (median imputation, unseen-category handling, dummy encoding, zero-variance and linear-combination filters) on the training data only.

Model	AUC	Recall	Precision	F1	Specificity	Balanced acc.	Verdict
Logistic regression (0.50)	0.759	0.026	0.441	0.049	0.992	0.509	Misses almost everyone
Naive Bayes (0.50)	0.741	0.792	0.302	0.437	0.575	0.683	Too many false alarms
Random forest (0.50)	0.787	0.061	0.549	0.110	0.988	0.525	Good ranking, wrong cut-off
Random forest at 0.20 (final)	0.787	0.800	0.338	0.475	0.636	0.718	Selected

Table 5. Test-set comparison (n = 51,114). AUC 95% bootstrap interval for the final model: 0.783 to 0.791.

The accuracy trap. At the default 0.50 cut-off three of the four candidates scored about 81% accuracy but found only 3% to 7% of unemployed people (the bootstrap interval for the forest's recall is 0.056 to 0.066). Predicting "no" almost always looks right when 81% of records are not unemployed. So we judged models on recall, precision, specificity, balanced accuracy and ROC-AUC together. AUC values of 0.74 to 0.79 showed real signal that the default cut-off wasted. We swept 19 thresholds and picked 0.20 for the random forest (Figure 5): recall rises from 6% to 80% while precision falls from 55% to 34%.

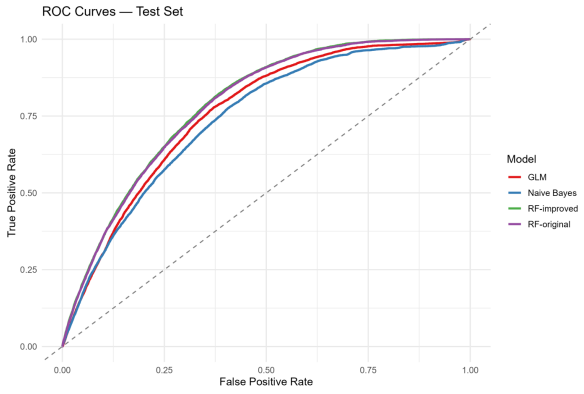


Figure 4a. ROC curves on the test set (Member 2).

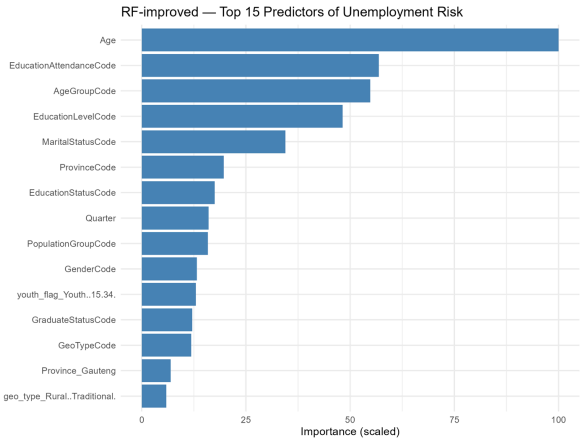


Figure 4b. Top 15 predictors, improved forest (Member 2).

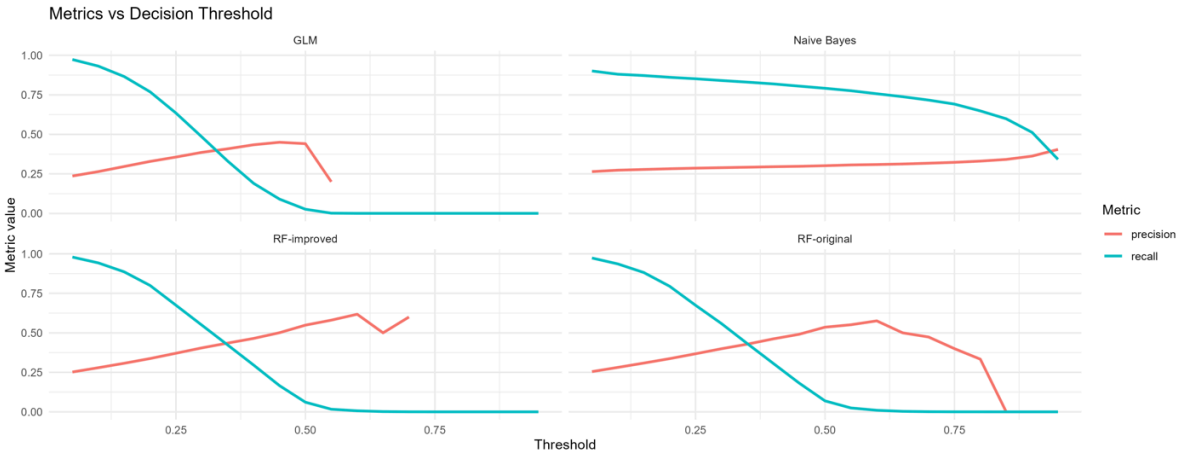


Figure 5. Precision and recall against decision threshold for each model .

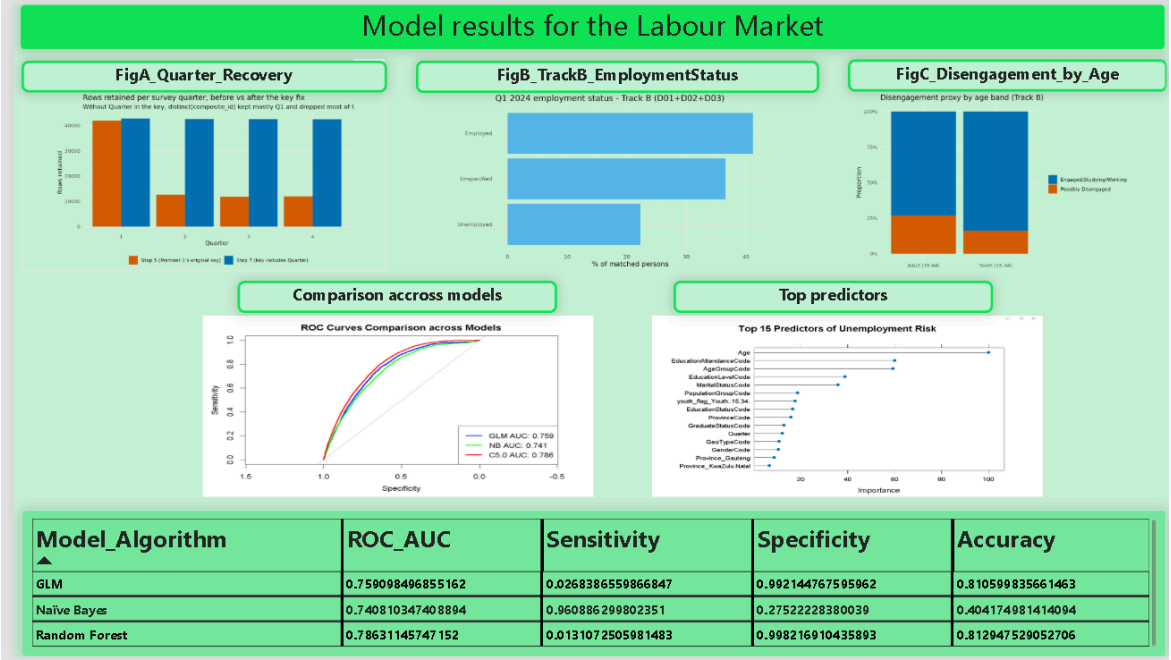


Figure 6. Dashboard Models page: model benchmarks and importance.

Reading the result honestly. At a threshold of 0.20 the model catches four in five unemployed people, but only about one in three flagged people is truly unemployed, and roughly 45% of everyone scored is flagged. It is a broad screening aid, not a precise targeting tool, and that is a deliberate trade-off: for a support programme, missing someone who needs help costs more than briefly reviewing someone who does not. Naive Bayes had similar recall but a specificity of 57.5% against 63.6%, so we rejected it. Age, school attendance, age group, education level, and marital status were the strongest predictors. These are predictive associations, not causes.

Who the model serves less well. Recall is 91% for young people but 64% for those 35 and over, and almost identical for women and men (79.9% and 80.1%). The serious gaps appear by population group (Black African 83%, Coloured 62%, Indian/Asian 37%, White 36%) and by area (rural traditional 82%, rural farms 50%). By province, it ranges from 66% in the Western Cape to 87% in the Free State. The gaps mostly reflect that groups with lower unemployment supply fewer positive examples to learn from. Recall was stable across quarters (78% to 83%). We judge the model fit for a supervised pilot as decision support, with the gaps disclosed, and not for automated decisions about individuals.

5. Deployment and monitoring

Tool choice. We compared R/Shiny, a plumber API, Power BI, and Quarto dashboard. Shiny won because it loads the exact saved recipe and model, so deployed scores match the validated ones, and it needs no extra infrastructure. A plumber API is the next step if another system needs to call the model. Power BI is used for reporting, not scoring, because it cannot run the ranger model natively.

The app. The SDHSA Unemployment Risk Explorer lets an analyst score one person or upload a CSV. Every input is validated first: required columns, numeric types, impossible ages, unseen categories and over-large files produce plain messages. The saved recipe and imputer then transform the data without refitting, the forest returns a probability, and a record is flagged at 0.20 or above. Each result is framed as decision support: it shows the score, the threshold, and a banner saying the score is not a determination of employment status (Figure 7). Uploaded data is not stored. A smoke test confirms that the deployed pipeline reproduces the baked test set used in evaluation, and checks the missing ages, unseen provinces, and empty uploads.

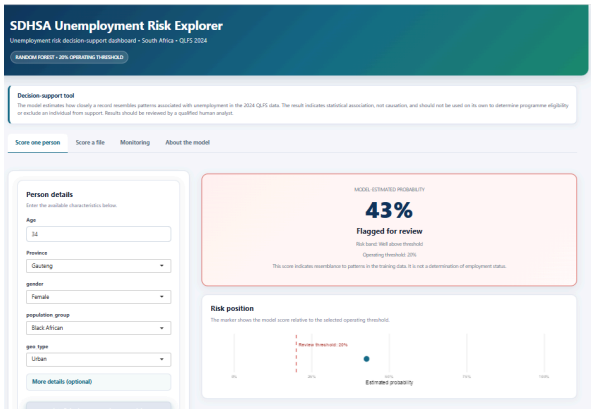


Figure 7. Scoring one person: 43% estimated probability, flagged for review at the 20% threshold.

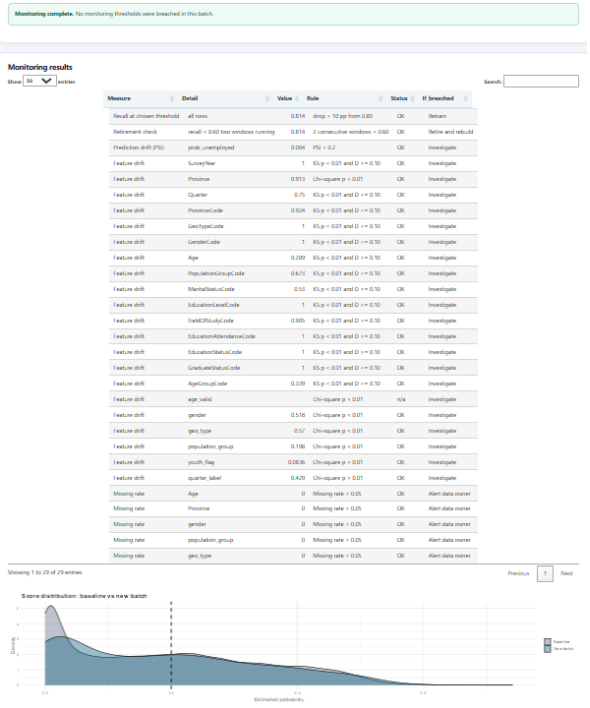


Figure 8. Monitoring results for an uploaded batch, with the run history.

Monitoring and retirement. A Monitoring tab runs the checks in Table 6 on each new batch and logs the run. The reference distribution comes from the held-out test set, not the training set, because the forest is over-confident on rows it was trained on and every new batch would look like drift. We tested it twice. A 29-row upload showed every applicable check as OK (the age validity check returned n/a). The saved log for a 500-row batch shows a score drift (PSI) of 0.004, no actions and recall of 0.814, above the 0.70 review trigger. A 29-row batch only proves the interface works: feature-drift tests need at least 500 rows and recall needs at least 30 known positives. Recall and fairness checks also need real outcomes, which arrive only with a new labelled QLFS release. Between releases the app can only watch inputs and score distributions. That is an early warning, not proof of accuracy, and the app does not retrain itself.

Check	Threshold	Action
Score drift (PSI)	above 0.20 (0.10 is a warning)	Investigate source data and coding
Feature drift (KS or chi-square)	p below 0.01 and a meaningful effect	Investigate; with large batches p-values alone flag everything
Missing data on key fields	above 5%	Alert data owner
Recall (needs outcomes)	drops more than 10 points from 0.80 (below 0.70)	Review and retrain
Population-group recall gap	widens by more than 10 points	Retrain with reweighting
Retirement	recall below 0.60 in two consecutive windows, or the QLFS instrument changes	Retire and rebuild

Table 6. Monitoring measures. The model is also reviewed at least yearly as new QLFS waves become available.

6. Ethics and responsible AI across the lifecycle

We treated ethics as a set of decisions made at each phase rather than a section written at the end.

Phase	Issue	What we did
Business understanding	Describing vulnerable groups can stigmatise them	Framed the work as finding where support is needed; neutral wording; association not cause
Data understanding	Re-identification from small cells; sensitive hardship data	Data is de-identified; small groups flagged; hardship files (D07, D09) left out
Preparation	Choices can quietly change who is counted (weights, deletion, recoding)	Kept legitimate extreme values; documented every recode with verification; weighted headline rates
Modelling	Protected attributes as inputs; imbalance	Kept gender and population group so fairness could be measured; outputs advisory; no resampling that hides imbalance
Evaluation	Accuracy hid a failure on the minority class; unequal group performance	Reported recall and precision by six dimensions and bootstrap intervals; disclosed gaps up to 46 points
Dashboard	Sample counts read as national totals; one province filter silently narrows every number	Sample and weighted views reconciled (Table 4); active filters visible; small groups flagged
Deployment	Misuse for eligibility or exclusion; personal data in uploads	Banner and model card rule out automated decisions; threshold and caveats shown; uploads not stored; hosting must comply with POPIA
Monitoring	Silent decay or widening fairness gaps	Batch checks, group-gap trigger, logged runs, retirement rule

Table 7. Ethical issues and mitigations by CRISP-DM phase.

Bias. The main source is structural: the data reflects unequal labour-market outcomes, and a model trained on it learns that pattern. Smaller groups give noisier results, and base-rate differences cause lower recall for groups with less unemployment. None of this can be removed from observational data, so the safeguards are transparency, human review and the monitoring triggers. We chose not to set different thresholds by race, which would be hard to defend in practice; any adjustment should be made at programme level with DEL.

Transparency and AI use. The threshold, its reason and the subgroup results are in the app, the model card and this report. We used AI tools for editing, code, debugging, and checking for errors. The full log is in the ethics report. The team checked and ran the outputs and takes responsibility for them.

7. Recommendations

Each recommendation follows from a result above.

- **Make young people the first priority.** They are 59.5% of unemployed people and have double the adult rate, and 42% of them are NEET. Fast-track workplace readiness and test youth wage subsidies rather than broad, untargeted job creation. Young women (49.4%) and 15 to 24 year olds (60.1%) need the most attention, so build gender-responsive support into youth programmes.

- **Use rates and headcounts together to place resources.** Target intensive programmes where rates are highest (North West, Eastern Cape, Free State, rural traditional areas) and scale volume-based programmes where most people are (Gauteng, KwaZulu-Natal) before allocating flat grants nationwide.

- **Look at the barriers in rural traditional areas.** Their rate (42.9%) and disengagement are the highest. Transport, connectivity and market access are plausible barriers and worth investigating, but our data cannot show they are the cause.

- **Focus on the school-to-work transition.** Two thirds of unemployed people hold a secondary or matric qualification, and their rate matches that of people with less schooling. Placement, work experience and bridging programmes for this group are likely to reach the most people.

- **Use the screening model as a first filter, not a verdict.** Use it to prioritise outreach lists and review flagged records by hand, expecting two in three flags to be wrong at 0.20. If review capacity is limited, test a higher threshold (Figure 5) before launch.

- **Do not rely on the model for White, Indian/Asian or farm-area residents**. Recall there is 36% to 50%. Use other identification routes, such as community or employer referrals.

- **Adopt the monitoring plan from day one**. Run the batch checks every quarter, review the model yearly, and have DEL decide who owns the monitoring log.

8. Limitations, uncertainty and future work

- **Associations, not causes.** Observational survey data cannot show why people are unemployed.
- **One year, random split.** Everything is from 2024, and a random split does not show how the model performs on a future quarter. Quarter and its derived fields may encode patterns specific to 2024, and province and quarter appear in more than one coded form. Temporal validation and a review of these redundant predictors come first.
- **Target definition.** The model predicts unemployed against a combined group of employed, inactive and discouraged people. It does not keep the four-way status.
- **Uncertainty and precision.** The AUC interval is narrow (0.783 to 0.791), but results for small groups (Indian/Asian 3,450 records, farms 6,240) are less certain. At 0.20 about two in three flags are false alarms and about 45% of people are flagged.
- **Parallel tracks and sectors.** Track B could not be linked to Track A, so education and job-search detail from D02 and D03 is not in the model, and the formal/informal coding (Section 2) leaves Objective 4 open.
- **Dashboard and deployment.** The dashboard shows sample counts, not weighted ones, and has not been tested with DEL staff. The app has no login, has not been tested with real DEL users,

and does not retrain automatically. Hosting personal data needs POPIA-compliant arrangements.

Future work. Add later QLFS waves to separate trends from one-off changes and test on a held-out future quarter; add weighted measures to the dashboard; resolve the formal/informal coding so sectors can be analysed; test group-aware retraining or reweighting for low-recall groups; test thresholds with programme staff; add a plumber API; and run a short usability test with DEL analysts.

9. Conclusion

Young people, residents of rural traditional areas and people in North West, Eastern Cape and Free State face the highest unemployment risk, while Gauteng and KwaZulu-Natal hold the largest numbers of people affected. A weighted analysis of 170,377 survey records reproduces the official quarterly rates, which gives the group comparisons weight, and the Power BI dashboard shows the same patterns in the sample. A random forest screening model finds 80% of unemployed people at a threshold of 0.20, at the cost of many false alarms and clearly unequal performance across groups. Used as supervised decision support with the monitoring plan in place, the dashboard and the app give DEL a practical way to decide where to look first. They do not explain unemployment, and they should not make decisions about individuals.

References

Chapman, P., Clinton, J., Kerber, R., Khabaza, T., Reinartz, T., Shearer, C. and Wirth, R. (2000) CRISP-DM 1.0: Step-by-step data mining guide. SPSS Inc.

Statistics South Africa (2024a) Quarterly Labour Force Survey, Quarter 1: 2024. Pretoria: Stats SA. Available at: https://www.sanews.gov.za/south-africa/sa-unemployment-rate-increases (media coverage of the release).

Statistics South Africa (2024b) Quarterly Labour Force Survey, Quarter 2: 2024. Media release. Available at: https://statssa.gov.za/publications/P0211/Media release QLFS Q2 2024.pdf

Statistics South Africa (2024c) Quarterly Labour Force Survey, Quarter 3: 2024. Media release. Available at: https://statssa.gov.za/publications/P0211/Media Release QLFS Q3 2024.pdf

Statistics South Africa (2024d) Quarterly Labour Force Survey, Quarter 4: 2024. Media release. Available at: https://statssa.gov.za/publications/P0211/Media%20Release%20QLFS%20Q4%202024.pdf

Protection of Personal Information Act 4 of 2013 (South Africa).

Appendix A. Final data dictionary

Analytical dataset used for the models and the dashboard (Track A, 170,377 rows). Full dictionaries for both tracks are in the repository (data/Cleaned/Data_Dictionary_Integration.csv).

Variable	Source	Description
record_id	D11 keys	HouseholdID_PersonNumber_Year_Quarter; unique per row
quarter / quarter_label	D11 Quarter	1 to 4; label such as "Q1 2024"
Province	D11 Province	Standardised to nine provinces (spacing and case fixed)
gender, population_group, geo_type	D11 codes	Recoded labels: Male/Female; Black African, Coloured, Indian/Asian, White; Urban, Rural (Traditional), Rural (Farms)
age, age_band, youth_flag	D11 Age	Age 15-64; bands 15-24 to 55-64; Youth (15-34) vs Non-Youth (35+)
edu_group	D11 EducationLevelCode	Primary or lower (<10); Secondary / Matric (10-12); Tertiary (13-29); Other / unspecified (98, 99, missing)
status_label	D11 EmploymentStatusCode	Employed, Unemployed, Discouraged Seeker, Other Inactive
employed, unemployed, discouraged, in_labour_force	Derived	0/1 indicators used for weighted rates
neet	D11 NEETCode	Not in education, employment or training: Yes/No
monthly_earnings, hours_worked	D11	Employed people only; sentinel and negative values set to missing; descriptive use only (leakage risk as model inputs)
quarter_weight	D11 QuarterWeight	Survey weight: number of people each record represents. Used for all headline rates
Unemployed (model target)	Derived	1 = Unemployed; 0 = Employed, Discouraged Seeker or Other Inactive

Appendix B. Repository, reproduction and team contributions

Run order from the project root (full instructions in the handover document):

Script	Purpose (CRISP-DM stage)
Milestone 1 notebook	Business and data understanding, EDA, data quality
Milestone 2 notebook	Data preparation: cleaning, integration, features, recipe, split
02b_fix_baked.R	Preparation fix: standardise Province, re-bake train and test
Milestone3.Rmd, 03_evaluation.R, 03b_rf_balanced.R, 03c_finalise.R	Modelling, evaluation, final model and metadata
deployment/ (app.R, R/*, tests/)	Deployment: Shiny app, scoring, monitoring, smoke test
04_build_powerbi_data.R	Builds weighted Power BI tables from the cleaned panel
Dashboard.pbix	Power BI dashboard (Member 1)

Packages: tidyverse, recipes, rsample, caret, ranger, e1071, pROC, shiny, DT, ggplot2, dplyr, stringr. Seed 2026 is used throughout.

Member	Main contribution to Milestone 4
Member 1: Power BI and insights	Final dashboard (.pbix), screenshots, findings and dashboard design sections, dashboard demonstration
Member 2: data, R and modelling	Data preparation and integration evidence, modelling and evaluation, Shiny app and monitoring, R scripts and data dictionary
Member 3: report, ethics and integration	Final report and CRISP-DM structure, business problem and methodology, ethics report and AI log, limitations and recommendations, presentation, repository and handover notes